

AI Usage DISCLOSURE FORM

1. Team Details

Team Name: The Defective Pieces

Project / Product Name: ReceiptVault

Organization / Institution (if any): MS Ramaiah Institute of Technology

Submission Date: May 8, 2026

2. AI Usage Declaration

Did your team use any Artificial Intelligence (AI) in developing this project? ☒ Yes

AI tools used: Google Gemini 2.5 Flash (vision & extraction), Claude (orchestration architecture & code assistance)

3. Purpose of AI Usage (Brief Details)

Idea generation / brainstorming Used Claude to design the agentic skill architecture, orchestration flow, and agent personality (SOUL.md)

Code generation or assistance Claude assisted with TypeScript service scaffolding, BullMQ queue setup, Supabase integration, and pipeline utilities

UI / UX design Claude assisted with component structure for the React Native / Expo frontend (GlowCard, BottomSheet, SkeletonLoader, etc.)

Content creation N/A — no AI-generated marketing or written content used

Data analysis Gemini 2.5 Flash extracts structured item-level data (store, items, totals, return windows, warranties) from receipt images

Testing / debugging Claude used to debug pipeline issues (receipt validation schema, BullMQ job scheduling, Gmail OAuth token flow)

Other Agentic orchestration: OpenClaw layer uses Claude-style reasoning to prioritize alerts and coordinate all 5 autonomous skills as a unified workflow

4. Feature Origin Classification (Please add/delete based on the number of the feature)

1. Feature Name: Receipt Scanning & Data Extraction

Origin: AI-Generated

Description: Tool: Google Gemini 2.5 Flash (geminiVision.ts). Prompt: Structured JSON extraction from receipt image (store, items, totals, return window, warranty). Output: Validated JSON stored in Supabase. Modification: Added Zod schema validation layer on top of raw AI output.

2. Feature Name: Return Deadline Tracker (deadline-watch skill)

Origin: Both

Description: Tool: Gemini 2.5 Flash (extracts return_deadline_days from receipt). Self-built: BullMQ job scheduler that queues alerts N days before expiry. Output: WhatsApp reminder sent via Meta Cloud API. Modification: Added quiet-hours and rate-limit guardrails.

3. Feature Name: Subscription Manager (sub-manager skill)

Origin: Both

Description: Tool: Claude (architecture design). Self-built: Detection logic for recurring charges in receipt/email history. Output: Renewal alerts with KEEP/CANCEL reply prompts. Modification: Confirmation flow before recording cancellation intent.

4. Feature Name: Consumable Tracker (consumable-tracker skill)

Origin: Both

Description: Tool: Gemini 2.5 Flash (is_consumable flag per item). Self-built: Purchase interval prediction using memory/patterns.yaml. Output: Reorder suggestion with Amazon search link. Modification: Per-user preference tuning via prefs.yaml.

5. Feature Name: Gmail Order Email Scanner (gmailScanner.ts)

Origin: Both

Description: Tool: Gemini 2.5 Flash (parses email body for order details). Self-built: Gmail OAuth2 integration scanning Amazon, Flipkart, Zomato, Swiggy, BigBasket. Output: Order data ingested into the same receipt pipeline. Modification: Duplicate-detection before insert.

6. Feature Name: OpenClaw Orchestration Layer

Origin: Both

Description: Tool: Claude (used to design agent rules, priority logic, SOUL.md spec). Self-built: Scheduler (scheduler.ts) runs all 5 skills as daily cron jobs 8:00–8:20 AM. Output: Unified agentic workflow with rate limits (max 3 alerts/hr/user) and action prioritization (return > subscription > consumable).

5. Ethical & Compliance Confirmation

AI usage complies with guidelines and policies. **Yes**

No proprietary or copyrighted data misused. **I Agree**

6. Declaration & Sign-Off

Name of Team Representative: Sourav Kumar Sahu

Role: Developer

Signature: 

Date: May 8, 2026